ДИС консултация 2024г.

1. Основни производни и интеграли

$(x^\alpha)' = \frac{x^{\alpha}}{1} \cdot \alpha (x^{\alpha-1})$

$(e^x)' = e^x$

$(\ln(x))' = \frac{1}{x}$

$(\sin x)' = \cos x$

$(\cos x)' = -\sin x$

$(\operatorname{tg} x)' = \frac{1}{\cos^2 x}$

$(\operatorname{ctg} x)' = -\frac{1}{\sin^2 x}$

$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$

$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$

$(\operatorname{arctg} x)' = \frac{1}{1+x^2}$

$(\operatorname{arcctg} x)' = -\frac{1}{1+x^2}$

Основни интеграли:

$$\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} + C, \alpha \neq -1$$

$$\int e^x dx = e^x + C$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \frac{1}{\cos^2 x} dx = \operatorname{tg} x + C$$

$$\int \frac{1}{\sin^2 x} dx = -\operatorname{cotg} x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \begin{cases} \arcsin x + C \\ -\arccos x + C \end{cases}$$

$$\int \frac{1}{1+x^2} dx = \begin{cases} \operatorname{arctg} x + C \\ -\operatorname{arccotg} x + C \end{cases}$$

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln\left(x + \sqrt{x^2+a^2}\right) + C$$

Δ - интервал

$f : \Delta \to R$

Казваме, че $F : \Delta \to R$ е примитивна на f в Δ, ако F е диференцируема за $\forall x \in \Delta$ и $F'(x) = f(x)$ за $\forall x \in \Delta$.

14.07.2015г.

$$\int_0^{1/2} \arcsin x \, dx = \left| \begin{array}{l} \text{интегриране по части} \\ \int f(x) \, d\, g(x) = \\ f(x) \cdot g(x) - \int g(x) \, d\, f(x) \end{array} \right| =$$

$$= x \cdot \arcsin x - \int x \, d \arcsin x =$$

$$= x \cdot \arcsin x - \int \frac{x}{\sqrt{1-x^2}} dx =$$

$$= x \cdot \arcsin x - \frac{1}{2} \int \frac{d\, x^2}{\sqrt{1-x^2}} =$$

$$= x \cdot \arcsin x + \frac{1}{2} \int \frac{d(1-x^2)}{\sqrt{1-x^2}}$$

$$= x \cdot \arcsin x + \frac{1}{2} \int (1-x^2)^{-\frac{1}{2}} \, d(1-x^2)$$

$$= x \cdot \arcsin x + \frac{1}{2} \cdot \frac{(1-x^2)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C$$

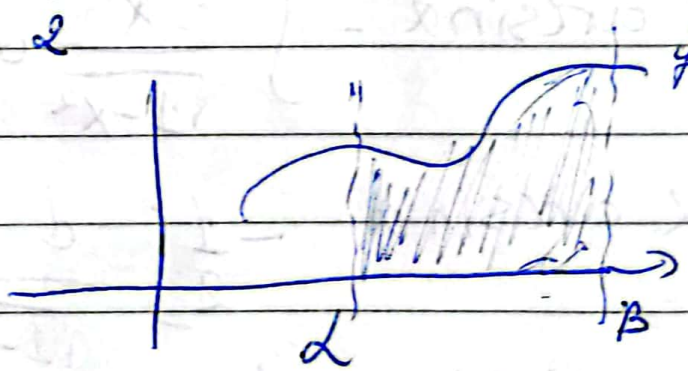
$$\int_0^{1/2} x \arcsin x \, dx =$$

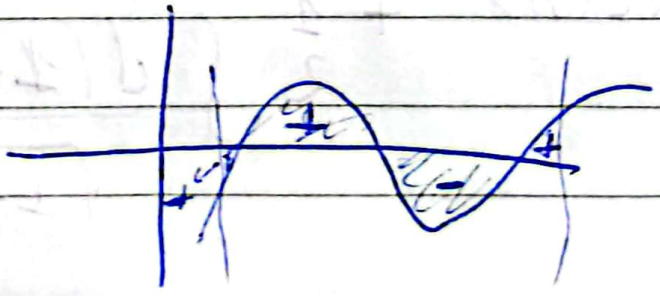
$$= x \cdot \arcsin x + \frac{1}{2} \cdot \frac{(1-x^2)^{\frac{1}{2}}}{\frac{1}{2}} \Bigg|_0^{\frac{1}{2}}$$

$$= \frac{1}{2} \cdot \arcsin \frac{1}{2} + \frac{1}{2} \cdot \frac{\left(1-\frac{1}{4}\right)^{\frac{1}{2}}}{\frac{1}{2}} - 0$$

$$= \frac{1}{2} \cdot \frac{\pi}{6} + \sqrt{\frac{3}{4}} - 1 = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$$

$$\int_2^3 f(x) \, dx$$





10.09.2015г

$$= \int \frac{x-2}{x(x^2+2)}\,dx$$

$$\frac{x-2}{x^2+(x^2+2)^2} =$$

$$= \frac{A}{x} + \frac{Bx2}{x2} + \frac{Cx+D}{x^2+2} + \frac{Ex+F}{(x^2+2)^2}$$

$$\frac{x-2}{x(x^2+2)}$$

$$\frac{A}{x} + \frac{Bx+C}{x^2+2} = \frac{x-2}{x(x^2+2)}$$

$$\frac{A(x^2+2) + (Bx+C)x}{x(x^2+2)} = \frac{x-2}{x(x^2+2)}$$

$$Ax^2 + 2A + Bx^2 + Cx = x-2$$

$$\begin{cases} A+B=0 \\ C=1 \\ A=-1 \\ B=1 \end{cases}$$

$$= \frac{x-2}{x(x^2+2)} = -\frac{1}{x} + \frac{x+1}{x^2+2}$$

$$I = \int -\frac{1}{x}\,dx + \underbrace{\int \frac{x+1}{x^2+2}\,dx}_{I_2} =$$

$$= -\ln|x| + C + I_2$$

$$I_2 = \int \frac{x}{x^2+2}\,dx + \int \frac{1}{x^2+2}\,dx$$

$$I_1 = \frac{1}{2}\int \frac{dx^2}{x^2+2} + \int \frac{1}{x^2+2}\,dx$$

$$I_1 = \ln(x^2+2) + \frac{1}{\sqrt{2}}\operatorname{arctg}\left(\frac{1}{\sqrt{2}}\right) + C$$

12.07.2016r.

$$\int_1^2 x \log_2 \frac{1}{x}\,dx$$

$$= \int_1^2 x \frac{\ln \frac{1}{x}}{\ln 2}\,dx = \left| \log_a b^p = p \log_a b \right.$$

$$= \frac{1}{\ln 2}\int_1^2 x \cdot \ln \frac{1}{x}\,dx$$

$$= \frac{1}{\ln 2}\int_1^2 x \cdot \ln x^{-1}\,dx =$$

$$= -\frac{1}{\ln 2}\int_1^2 x \cdot \ln x\,dx$$

$$= -\frac{1}{\ln 2}\,\frac{1}{2}\int_1^2 \ln x\,dx^2$$

$$= -\frac{1}{\ln 2}\,\frac{1}{2}\cdot\left(x^2 \ln x\Big|_1^2 - \int_0^1 x^2\,d\ln x\right)$$

$$= -\frac{1}{2\ln 2}\cdot\left(4\ln 2 - \ln 1 - \int_1^2 \frac{x^2}{x}\,dx\right)$$

$$= -\frac{1}{2\ln 2}\cdot\left(4\ln 2 - \ln 1 - \frac{x^2}{2}\Big|_1^2\right)$$

$$= -\frac{1}{2\ln 2}\left(4\ln 2 - \ln 1 - \frac{4}{2} + \frac{1}{2}\right)$$

$$= -2\ln 2 + \frac{1}{2}\frac{\ln 1}{\ln 2}$$

09.09.2016г.

$$I = \int_0^{\frac{\pi}{2}} x\cdot\cos^2 x\,dx$$

$$\cos^2 x = \frac{1+\cos 2x}{2}$$

$$\sin^2 x = \frac{1-\cos 2x}{2}$$

$$\cos 2x = 2\sin x\cdot\cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x$$

$$\cos 2x = \cos^2 x - (1-\sin^2 x)$$

$$I = \int_0^{\frac{\pi}{2}} x\cdot\left(\frac{1+\cos 2x}{2}\right)dx =$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} x\,dx + \frac{1}{2}\int_0^{\frac{\pi}{2}} x\cdot\cos 2x\,dx$$

$$= \frac{1}{2}\left(\frac{x^2}{2}\Big|_0^{\frac{\pi}{2}}\right) + \frac{1}{2}\cdot\frac{1}{2}\int_0^{\frac{\pi}{2}} x\cdot\cos 2x\, d2x$$

$$= \frac{1}{2}\left(\frac{\pi^2}{8} + \frac{1}{4}\int_0^{\frac{\pi}{2}} x\, d\sin 2x\right)$$

$$= \frac{1}{2}\left(\frac{\pi^2}{16} + \frac{1}{4}\left(x\cdot\sin 2x\Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}}\sin 2x\, dx\right)\right)$$

$$= \frac{\pi^2}{16} + \frac{1}{4}\left(0 - 0 - \frac{1}{2}\int_0^{\frac{\pi}{2}}\sin 2x\, d2x\right)$$

$$= \frac{\pi^2}{16} - \frac{1}{8}\left(-\cos 2x\Big|_0^{\frac{\pi}{2}}\right) =$$

$$= \frac{\pi^2}{16} - \frac{1}{8}\left(1 + 1\right)$$

$$= \frac{\pi^2}{16} + \frac{1}{8}\cos 2x\Big|_0^{\frac{\pi}{2}} =$$

$$= \frac{\pi^2}{16} + \frac{1}{8}\left(-1 - 1\right) = \frac{\pi^2}{16} - \frac{1}{4}$$

$$\int_1^2 \frac{x+1}{\sqrt{x}(1+\sqrt{x})}\,dx$$

$\sqrt{x} = t \geq 0$

$x = t^2$

$dx = dt^2 = 2t\,dt$

$$\Rightarrow \int_1^{\sqrt{2}} \frac{(t^2+1)\cdot(2t)}{t(1+t)}\,dt$$

$$= 2\int_1^{\sqrt{2}} \frac{t^2+1}{t+1}\,dt$$

$t^2+1 : t+1 = t-1$

t^2+t

$-t+1$

$-t+1$

2

$$\frac{t^2+1}{t+1} = t-1+\frac{2}{t+1}$$

$$2\int_1^{\sqrt{2}} \frac{t^2+1+1-1}{t+1}\,dt =$$

$$= 2\int_1^{\sqrt{2}} \frac{t^2-1}{t+1}\,dt + 2\int_1^{\sqrt{2}} \frac{2}{t+1}\,dt =$$

$$= 2\int_1^{\sqrt{2}} t-1\,dt + 2\ln(t+1)\Big|_1^{\sqrt{2}}$$

$$= 2\left(\frac{(t-1)^2}{2}\right)\Big|_1^{\sqrt{2}} + 4\ln(t+1)\Big|_1^{\sqrt{2}}$$

$$I = \int_0^{\frac{\pi}{2}} \left|\frac{1}{2} - \sin^2 x\right| dx$$

$$\int_0^{\frac{\pi}{2}} \left|\frac{1}{2} - \left(\frac{1-\cos 2x}{2}\right)\right| dx$$

$$= \int_0^{\frac{\pi}{2}} \left|\frac{1}{2} - \frac{1}{2} + \frac{\cos 2x}{2}\right| dx =$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} |\cos 2x|\,dx$$

$$|\cos 2x| = \begin{cases} \cos 2x, & x \in \left(0, \frac{\pi}{4}\right] \\ -\cos 2x, & x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \end{cases}$$

$$I = \frac{1}{2}\int_0^{\frac{\pi}{4}} \cos 2x\,dx + \frac{1}{2}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} -\cos 2x\,dx$$

$$I = \frac{1}{2}\cdot\frac{1}{2}\int_0^{\frac{\pi}{4}} \cos 2x\,d2x - \frac{1}{2}\cdot\frac{1}{2}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos 2x\,d2x$$

$$I = \frac{1}{4}\sin 2x\Big|_0^{\frac{\pi}{4}} - \frac{1}{4}\cdot\sin 2x\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \frac{1}{4}\cdot\left(\sin\frac{\pi}{2} - \sin 0\right) - \frac{1}{4}\cdot\left(\sin\pi - \sin\frac{\pi}{2}\right)$$

$$= \frac{1}{4}\cdot(1-0) - \frac{1}{4}\cdot(0-1) =$$

$$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \qquad \text{Ответ } \frac{1}{2}$$

$$\int_0^{\frac{\pi}{2}} \frac{dx}{2+\sin x}$$

$$2+\sin x = 2 + 2\sin\frac{x}{2}\cdot\cos\frac{x}{2} =$$

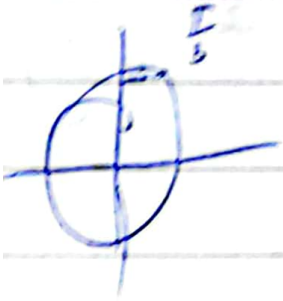
$$= 2\cdot\left(\sin^2\frac{x}{2} + \cos^2\frac{x}{2}\right) + 2\sin\frac{x}{2}\cdot\cos\frac{x}{2} \,/\cos^2\frac{x}{2}$$

Top chek

$$= 2\cos^2\frac{x}{2}\left(\operatorname{tg}^2\frac{x}{2} + \operatorname{tg}\frac{x}{2} + 1\right)$$

$$= \int_0^{\frac{\pi}{2}} \frac{2\,d\,2^{\frac{x}{2}}}{2\cos^2\frac{x}{2}\left(2\operatorname{tg}^2\frac{x}{2} + 2\operatorname{tg}\frac{x}{2} + 2\right)}$$

$$= \int_0^{\frac{\pi}{2}} \frac{d\operatorname{tg}\frac{x}{2}}{\operatorname{tg}^2\frac{x}{2} + \operatorname{tg}\frac{x}{2} + 1}$$



$$\int \frac{d\operatorname{tg}\frac{x}{2}}{\operatorname{tg}^2\frac{x}{2} + \operatorname{tg}\frac{x}{2} + 1} \qquad \operatorname{tg}\frac{x}{2} = y$$

$$\int \frac{dy}{y^2 + y + 1} =$$

$$= \int \frac{dy}{\left(y + \frac{1}{2}\right)^2 + \frac{3}{4}} =$$

$$= \int \frac{d\left(y + \frac{1}{2}\right)}{\left(y + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} =$$

$$= \int \frac{d\left(y + \frac{1}{2}\right)}{\frac{3}{4}\cdot\left(\frac{4}{3}\left(y + \frac{1}{2}\right)^2 + 1\right)} =$$

$$= \frac{4}{3}\int \frac{\frac{2}{\sqrt{3}}\,d\left(\frac{2}{\sqrt{3}}y\right)}{\left(\frac{2}{\sqrt{3}}y + \frac{1}{\sqrt{3}}\right) + 1} =$$

$$= \frac{2\sqrt{3}}{3} \operatorname{arctg}\left(\frac{2y+1}{\sqrt{3}}\right) + C$$

$$F(x) = \frac{2\sqrt{3}}{3} \operatorname{arctg}\left(\frac{2\operatorname{tg}\frac{x}{2}+1}{\sqrt{3}}\right) + C$$

$$F\left(\frac{\pi}{2}\right) - F(0) = \frac{2\sqrt{3}}{3} \cdot \operatorname{arctg}\left(\frac{2\operatorname{tg}\frac{\pi}{4}+1}{\sqrt{3}}\right)$$

$$- \frac{2\sqrt{3}}{3} \cdot \operatorname{arctg}\left(\frac{2\operatorname{tg}0+1}{\sqrt{3}}\right) =$$

$$= \frac{2\sqrt{3}}{3} \left| \operatorname{arctg}\sqrt{3} - \operatorname{arctg}\frac{1}{\sqrt{3}} \right. = \frac{2\sqrt{3}}{3} \left| \frac{\pi}{3} - \frac{\pi}{6} \right.$$

10.09.2018г.

$$\int_0^1 x^2 \operatorname{arctg} x \, dx =$$

$$= \frac{1}{3} \int_0^1 \operatorname{arctg} x \, d\frac{x^3}{3} =$$

$$= \frac{1}{3} \operatorname{arctg} x \cdot \frac{x^3}{3} \Big|_0^1 - \frac{1}{3} \int x^3 \, d \operatorname{arctg} x$$

$$= \frac{1}{3} \cdot \frac{\pi}{4} - \frac{1}{3} \int_0^{\pi/2} \frac{x^3}{1+x^2} \, dx$$

$$= \frac{\pi}{12} - \frac{1}{3} \int_0^{\frac{\pi}{2}} \frac{x^3 + x - x}{1+x^2} \, dx$$

$\int x \operatorname{arctg} \sqrt{x} \, dx$

podstawienie

$\sqrt{x} = y \geq 0$

$x = y^2$

$\int y^2 \operatorname{arctg} y \, dy$

$$= \frac{\pi}{12} - \frac{1}{3}\int_0^{\frac{\pi}{4}} \frac{x^3+x}{x^2+1}dx + \frac{1}{3}\int_0^{\frac{\pi}{4}} \frac{1}{x^2+1}dx =$$

$$= \frac{\pi}{12} - \frac{1}{3}\int_0^{\frac{\pi}{4}} \frac{x\,dx}{x^2+1} + \frac{1}{6}\int_0^{\frac{\pi}{4}} \frac{dx^2+1}{x^2+1}$$

$$= \frac{\pi}{12} - \frac{1}{3}\frac{x^2}{2}\Big|_0^1 + \frac{1}{6}\ln(x^2+1)\Big|_0^1 =$$

$$= \frac{\pi}{12} - \frac{1}{3}\cdot\frac{1}{2} + \frac{1}{6}\cdot\ln 2 = \frac{\pi}{12} - \frac{1}{6} + \frac{1}{6}\ln 2$$

09.07.2019г

$$I = \int_1^8 \frac{dx}{\sqrt[3]{x}+x} =$$

$\sqrt[3]{x} = y$; $x = y^3$ $dx = dy^3 =$

$3y^2dy$

$$\Rightarrow I = 3\int_1^2 \frac{3y^2dy}{y^3+y} = 3\int_1^2 \frac{3y}{y^2+1}dy$$

$$= \frac{3}{2}\ln(y^2+1)\Big|_1^2$$

10.09.2019г

$$\int_0^2 \ln(x^2+4)\,dx =$$

$$= x\cdot\ln(x^2+4)\Big|_0^2 - \int_0^2 x\,d\ln(x^2+4)$$

$$= 2\ln 8 - \int_0^2 \frac{x\cdot 2x}{x^2+4}\,dx$$

$$= 2\ln 8 - 2\int_0^2 \frac{x^2+4-4}{x^2+4}\,dx =$$

$$= 2\ln 8 - 2\int_0^2 dx + 8\int_0^2 \frac{dx}{x^2+4} =$$

$$= 2\ln 8 - 2x\Big|_0^2 + 8\cdot\frac{1}{2}\operatorname{arctg}\frac{x}{2}\Big|_0^2$$

$$= 2\ln 8 - 4 + 4(\operatorname{arctg} 1 - \operatorname{arctg} 0)$$

$$= 2\ln 8 - 4 + 4\left(\frac{\pi}{4} - 0\right) = 2\ln 8 - 4 + \pi$$

08.09.2021г.

$$\int \operatorname{arctg}\sqrt{x}\,dx \quad \underline{y=\sqrt{x}>0} \quad 2\int \operatorname{arctg} y\,dy$$

$$= 2\cdot\frac{1}{2}\int \operatorname{arctg} y\,dy^2 \ldots$$

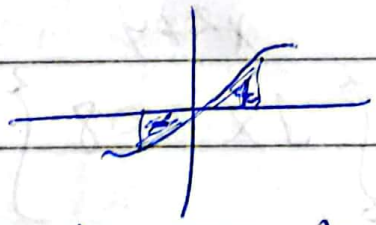
13.07.2021г.

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{|x|}(\sin x + \cos x)\,dx$$

$f: [-a, a] \to \mathbb{R}$

$f(-x) = -f(x)$ (нечетна)

$$\Rightarrow \int_{-a}^{a} f(x)\,dx = 0$$



$f(-x) = f(x)$

$$\Rightarrow \int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{|x|}\sin x\,dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{|x|}\cdot(0)\,x$$

нечетна — четна

$$e^{|-x|}\cdot\sin(-x) = e^{|x|}\cdot -(\sin x) = -e^{|x|}\sin$$

$$\Rightarrow I = 0 + 2\int_{0}^{\frac{\pi}{2}} e^{x}\cdot\cos x\,dx$$

$$= 2\int_0^{\frac{\pi}{2}} e^x \cdot \cos x \, dx$$

$$J = \int e^x \cos x \, dx = \int \cos x \, de^x$$

$$= e^x \cdot \cos x - \int e^x \, d\cos x =$$

$$= e^x \cdot \cos x + \int \sin x \, e^x \, dx =$$

$$= e^x \cdot \cos x + \int \sin x \cdot de^x =$$

$$= e^x \cdot \cos x + e^x \cdot \sin x - \int e^x \cdot d\sin x$$

$$= e^x \cdot (\cos x + \sin x) + \underbrace{\int \cos x \cdot e^x \cdot dx}_{J}$$

$$\Rightarrow 2J = e^x (\cos x + \sin x)$$

$$J = \frac{e^x (\cos x + \sin x)}{2}$$

$$I = 2 \cdot \frac{1}{2} e^x (\cos x + \sin x) \Big|_0^{\frac{\pi}{2}} =$$

$$= e^{\frac{\pi}{2}} \cdot \left(\cos\frac{\pi}{2} + \sin\frac{\pi}{2}\right) - e^0 (\cos 0 + \sin 0)$$

$$= e^{\frac{\pi}{2}} (1) - 1(1) = e^{\frac{\pi}{2}} - 1$$

12.07.2022г.

$$I = \int_0^2 \sqrt{x}\ln(\sqrt{x}+1)$$

$$= \frac{1}{2}\int_0^1 \sqrt{x}\ln(x+1)\,dx$$

$$\sqrt{x} = y \qquad x = y^2$$

$$dx = dy^2 = 2y\,dy$$

$$\Rightarrow I = \frac{1}{2}\int_0^1 2y^2\ln(1+y^2)\,y\,dy =$$

$$= \int_0^1 y^2\ln(1+y^2)\,dy =$$

$$= \frac{1}{3}\int_0^1 \ln(1+y^2)\,dy^3 =$$

$$= \frac{1}{3}y^3\ln(1+y^2)\Big|_0^1 - \int y^3\,d\ln(1+y^2)$$

$$= \frac{1}{3}\cdot(\ln 2) - \frac{1}{3}\int \frac{y^3}{y^2+1}\cdot 2y\,dy$$

$$= \frac{1}{3}\ln 2 - \frac{2}{3}\int \frac{y^4}{y^2+1}\,dy =$$

$$= \frac{1}{3}\ln 2 - \frac{2}{3}\int \frac{y^4-1+1}{y^2+1}\,dy =$$

$= \frac{\ln 2}{3} - \frac{2}{3} \int \frac{y^2-1}{y^2+1} dy - \frac{2}{3} \int \frac{y^2+1}{y^2+1} dy =$

$= \frac{\ln 2}{3} - \frac{2}{3} \int y^2$

$= \frac{\ln 2}{3} - \frac{2}{3} \int y^2 - 1 \, dy - \frac{2}{3} \int \frac{dy}{y^2+1}$

$= \frac{\ln 2}{3} - \frac{2}{3} \left(\frac{y^3}{3} - y \right)$

07.07.2022.

$\int (x + \operatorname{tg}^2 x) \sin^2 x \, dx =$

$= \underbrace{\int x \cdot \sin^2 x \, dx}_{J_1} + \underbrace{\int \operatorname{tg}^2 x \cdot \sin^2 x \, dx}_{J_2}$

$J_1 = \int x \cdot \left(\frac{1 - \cos 2x}{2} \right) dx =$

$= \int \frac{x}{2} dx - \frac{x \cdot \cos 2x}{2} dx =$

$= \frac{1}{2} \cdot \frac{x^2}{2} + C - \frac{1}{2} \int x \cdot \cos 2x \, dx$

$$\int tg^2x \cdot \sin^2x\,dx =$$

$\frac{\sin^2x}{\cos^2x}$

$$= \int tg^2x(1-\cos^2x)\,dx =$$

$$= \int tg^2x\,dx - \int \frac{\sin^2x}{\cos^2x}\cdot \cos^2x\,dx$$

$$= \int tg^2x\,dx - \int \sin^2x\,dx$$

$tgx = y$

$$= \int \frac{(1-\cos^2x)^2}{\cos^2x} =$$

$$= \int \frac{1-2\cos^2x+\cos^4x}{\cos^2x}\,dx =$$

$$= \int \frac{1}{\cos^2x}\,dx - 2\int dx + \int \cos^2x\,dx$$

$$= tgx + C - 2x + C + \int \frac{1+\cos 2x}{2}\,dx$$

$$\int \frac{dx}{x^2-1}$$

$$\int \sqrt{a^2+x^2}\,dx$$

$$\int \sqrt{a^2-x^2}\,dx$$

$(x^2-a^2)^{\frac{1}{2}} = (x^2-a^2)^{-\frac{1}{2}} \cdot 2x$

$= \frac{x}{\sqrt{x^2-a^2}}$

$\int \frac{dx}{(x^2+1)^n}$

$I = \int \sqrt{x^2-a^2}\, dx = x\sqrt{x^2-a^2} - \int x \cdot \frac{x}{\sqrt{x^2-a^2}}$

$= x\sqrt{x^2-a^2} - \int x \cdot$

$// (x^2-a^2)^{\frac{1}{2}} = \frac{1}{2}(x^2-a^2)^{\frac{1}{2}-1} \cdot (2x^2-a^2)'$

$=$

$\Rightarrow x \cdot \sqrt{x^2-a^2} - \int \frac{x^2 - a^2 + a^2}{\sqrt{x^2-a^2}}\, dx =$

$= x \cdot \sqrt{x^2-a^2} - \int \sqrt{x^2-a^2}\, dx + a^2 \int \frac{dx}{\sqrt{x^2-a^2}} =$

$= x \cdot \sqrt{x^2-a^2} - \int \sqrt{x^2-a^2}\, dx + a^2 \ln|x - \sqrt{x^2-a^2}| + c$